

Figure 1: JavaFX user library will be created



Figure 2: JavaFX project with all the necessary files and JDKs attached

as in the current version of Visual Studio), in which users can select their required controls and add them to their respective work area, modify them and design them by applying CSS stylesheets. An FXML file gets generated automatically in the project you are working on, which is the essence of JavaFX.

After Java 8 update 40, Oracle has stopped providing installers for Scene Builder. They will be provided only as source code. But users need not worry, since an organisation named Gluon has taken up the task of helping the developer community by managing and updating the Scene Builder project. You only have to download it from the Gluon website <http://gluonhq.com/products/scene-builder/> free of cost, and the installer will install it on your machine.

Configuring JavaFX in Eclipse IDE

In this tutorial, I will use the Eclipse IDE version Neon (4.6) for compiling and running my code, but you can use any version above 3.4. Before we go into the actual coding process with JavaFX, let's see how we can install the JavaFX plugin in Eclipse because the latter doesn't provide support for JavaFX, by default. The requisite for this is that you should have Eclipse downloaded on your system and Java 8 installed, because we will use the latest JavaFX 8 to develop our application. If you don't have Eclipse, you can download it from <http://www.eclipse.org/downloads/> and you can also download Java 8 from Oracle's official website. Once downloaded, start Eclipse and set your workspace for your project. There are two ways in which you can provide support for JavaFX in your project. One is by adding a '.jar' file to your Java project and the other is by installing the JavaFX plugin. Going forward, we will explore both ways.

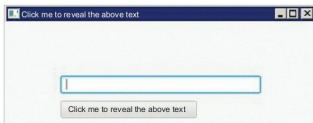


Figure 3: First JavaFX application



Figure 4: Output of the application

Adding a JavaFX jar

1. Create a new Java project in your workspace by going to *File menu > New* and giving the appropriate name.
2. Right-click on your created project and go to the *Properties* option.
3. Select *Java Build Path* in the *Properties* dialogue box.
4. Select the *Libraries* tab and check whether your JRE system library corresponds to Java version 8. If not, add it.
5. Now click on the *Add Library* button on the right side and select *User Library* from the dialogue box that appears. Click *Next*.
6. Click on the *User Libraries* button on the *Add Library* dialogue box and the *Preferences* window will open.
7. Click *New*, provide a suitable name for your library and then click *OK*.
8. Your library will be added to the libraries list as shown in Figure 1. Select that library and click on the *Add External JARs* button.
9. Now browse the folder where your Java 8 is installed, generally in your C: drive, and select the file *fxswt.jar* by going into the folder structure as `jdk 1.08_101 > jre > lib`.
10. Check the library that you have created and click *OK* or *Apply*, wherever applied.

After this, you can start developing your applications in JavaFX. Now, we will take a look at the second way in which we can associate JavaFX with Eclipse. By using this method, you can directly define a JavaFX project, unlike the first method in which we first defined a normal Java project and blended it into JavaFX using the JavaFX JARs.

Installing the JavaFX plugin

1. In the Eclipse IDE, click on the *Help* menu on the menu bar.
2. Under the *Help* menu, click on the option to *Install New Software*.
3. A dialogue box to install new software will open. In *Work with textbox*, type the URL <http://download.eclipse.org/efclipse/updates-released/2.4.0/site> and